INCH-POUND

MIL-PRF-21165B
13 January 2022
SUPERSEDING
MIL-A-21165A
27 August 1965

PERFORMANCE SPECIFICATION

ADAPTERS, QUICK-DISCONNECT, PASSENGER SEAT TO FLOOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for passenger seat to floor quick-disconnect adapters for installation in military aircraft.

1.2 Classification. The adapters are of the following types as specified (see 6.2).

TABLE I. Adapter part numbers and sizes.

Part Number	Diameter (inches)
MS22034-1	.855/ .850
MS22034-2	.885/ .882
MS22034-3	.867/.864
MS22034-4	.752/.748
MS22034-5	.893/.890
MS22034-6	.917/.914

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys
MS22034	-	Adapter, Quick-Disconnect, Passenger Seat to Floor

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-889	-	Galvanic Compatibility of Electrically Conductive Materials

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ H1331	-	Zero Acceptance Number Sampling Plans

(Copies of these documents are available from www.asq.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL

SAE AS21234	-	Adapter, Tiedown Cargo Ring and Seat Stud, 5,000 lb, Type I
SAE AS21235	-	Adapter, Tiedown Cargo Ring and Seat Stud, 10,000 lb, Type II

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(Copies of these documents are available from www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2. Materials. Materials shall be selected as to ensure compliance with the performance and finish requirements of this specification while minimizing weight.

3.2.1 Metal parts. Metal parts shall be of a corrosion-resistant material or treated in a manner to render them corrosion resistant.

3.2.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact.

3.2.1.2 Aluminum alloy parts. Aluminum alloy parts, when used, shall be anodized in accordance with MIL-PRF-8625 as applicable.

3.2.2 Other materials. When other materials are used in the construction of the adapters, they shall not deteriorate when exposed to environmental conditions likely to occur during service usage.

3.3 Design and construction. The adapters shall conform to MS22034. The adapters shall be designed and constructed to withstand the strains, jars, vibration, and other conditions incident to shipping, storage, installation, and service. The adapters shall not be subject to opening by shock or jar. It shall be designed to ensure smooth and positive operation between -65 °F and 350 °F.

3.4 Performance. The adapters shall conform to the following requirements when subjected to the applicable tests of 4.5.

3.4.1 Strength. The adapters tested shall not slip off the studs and shall not break when subjected to the loading specified in 4.5.2.1, 4.5.2.2, and 4.5.2.3, slight deformation being acceptable.

3.4.2 Vibration. The adapters shall not disengage when subjected to the vibration tests specified in 4.5.3.

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3.4.3 Endurance. The adapters shall be capable of sustaining 1000 cycles without malfunction when subjected to the endurance tests specified in 4.5.4.

3.5 Finish. Protective coatings and finishes that result in cracks, chips, or scaling during normal service life or under extremes of atmospheric conditions, shall not be used.

3.6 Identification of product. Each adapter shall be permanently marked at the location indicated on MS22034 for identification with the part number and the manufacturer's name symbol, or trademark in accordance with MIL-STD-130.

3.7 Workmanship. The adapters shall be free from irregularities or defects which could adversely affect performance, reliability, or durability (see 4.5.1).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, tests shall be performed at an ambient temperature of 70 °F ±20 °F.

4.3 First article inspection. First article inspection shall consist of all the tests and examinations specified in 4.3.1 and 4.4.

4.3.1 Testing requirements. After the award of the contract or order, the manufacturer shall submit twelve adapters of the MS part number contracted. After a visual examination, all adapters shall be subjected to the tests in 4.5.2, 4.5.3, and 4.5.4, followed by a final visual examination. The adapter samples shall be representative of the construction, workmanship, components, and materials to be used during production. Samples subjected to these tests shall not be delivered as part of the contract or order but shall be discarded.

4.4 Conformance inspection. The quality conformance inspection shall consist of the following:

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TABLE II. First article/conformance inspections.

Inspection	Requirement	Verification
Visual examination of the adapters	3.2 3.3 3.6 3.7	4.5.1
Strength tests	3.4.1	4.5.2 4.5.2.1 4.5.2.2 4.5.2.3
Vibration	3.4.2	4.5.3
Endurance	3.4.3	4.5.4

4.4.1 Sampling. Unless otherwise specified, sampling for inspection shall be performed in accordance with the provisions specified in ASQ H1331.

4.4.1.1 Inspection lot. For the purpose of determining the sample size in accordance with ASQ H1331, Critical, Major and Minor attributes shall be assigned Acceptance Quality Levels (AQLs) of 0.1, 1.0 and 4.0 respectively. Unspecified attributes shall be considered as Major, unless sampling plans are specified in applicable documents. The inspection lot size shall be expressed in units of adapters for examinations specified in 4.4.1.1.1. The adapters in the inspection lot shall be made under essentially the same conditions and from the same materials and components.

4.4.1.1.1 Examination of the end item. The sample unit shall be one adapter. The sample unit shall be visually inspected and measured to ensure it meets the requirements specified in 3.2 through 3.7.

4.5 Inspection methods.

4.5.1 Visual examination. Every adapter in a sample lot shall be checked dimensionally and examined visually to determine conformance to this specification and MS22034. The classification of defects (see table III) shall be used to classify the defects found.

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TABLE III. Classification of defects for visual examination of the adapter.

Major	Minor
101. Dimensions not within specified tolerances. 102. Surface – rough, malformed, misaligned or improperly machined. 103. Improper assembly – any component missing, malformed, fractured or otherwise damaged 104. Other discrepancies which might impair the function of the adapter. 105. Any functioning part that works with difficulty. 106. Any component loose or otherwise not securely retained. 107. Faulty workmanship which might impair the serviceability of the adapter.	201. Material – visual oxidation 202. Marking – missing, insufficient, incorrect, illegible, or not permanent.

4.5.2 Strength tests. Each sample adapter shall be subjected to the following strength tests and shall meet the requirements of 3.4.1.

4.5.2.1 Static horizontal loading. Apply a load of 2300 pounds to a collared adapter parallel to the base of the adapter, as shown on Figure 1.

4.5.2.2 Static vertical loading. Apply a load of 5475 pounds to a collared adapter through the longitudinal axis of the adapter, as shown on Figure 1.

4.5.2.3 Impact loading. A force of 5950 pounds resulting from impact loading shall be applied to the adapter at an angle of 22-1/2 degrees as shown on Figure 1. The impact loading shall be applied as a half-sine wave with a base of 0.050 second and a peak of 5950 pounds.

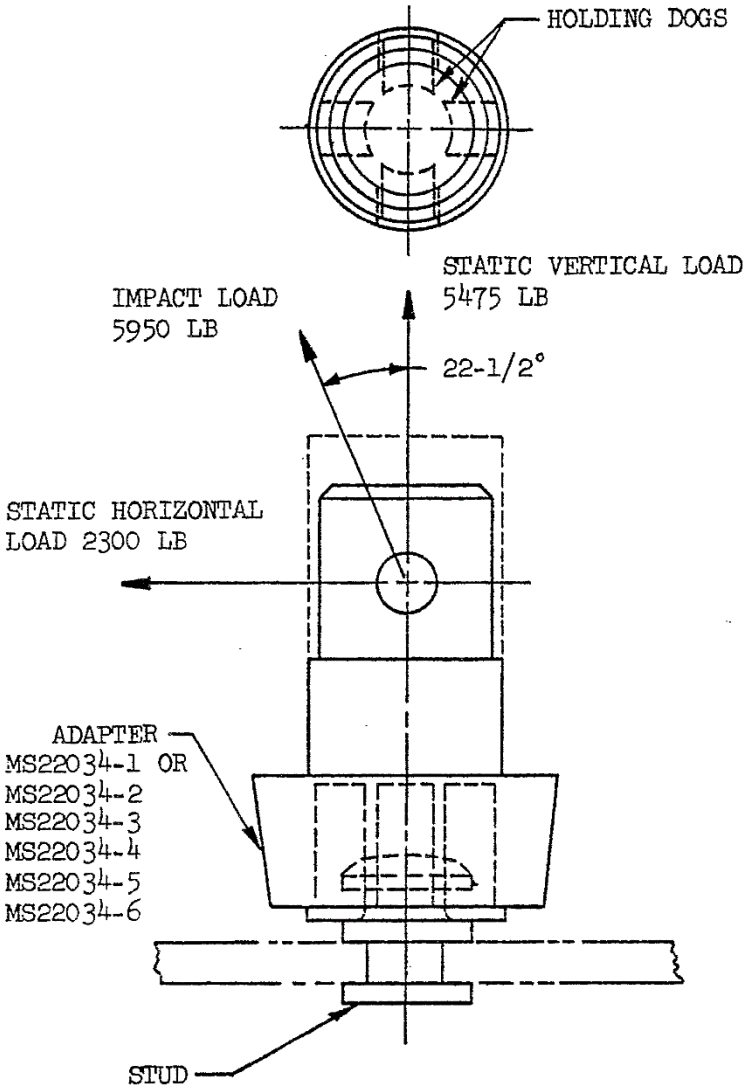
4.5.3 Vibration. Each sample adapter shall be fastened and vertically tensioned to 200 pounds and vibrated in this condition on a vibration table at a double amplitude of 1/16 inch at

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60 cycles per second for a period of 6 hours. The adapter shall meet the requirements of 3.4.2. The endurance test shall be conducted after the completion of this test.

4.5.4 Endurance. After being subjected to the vibration test, each adapter shall be subjected to 1000 cycles of engaging and disengaging. The adapter shall meet the requirements of 3.4.3.

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(SEE DETAIL A OF SAE AS21234 and SAE AS21235)

NOTE: The loading shall be applied to a collared adapter, the load acting through the pinhole to simulate actual loading transmitted by the leg attachment of the passenger seat.

FIGURE 1. Test load diagram.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The adapters covered by this specification are intended to provide a convenient means of connecting or disconnecting the passenger seat legs to the floor studs of cargo-transport type military aircraft.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Quantity desired
c. MS Part Number
d. Selection of applicable levels of packaging and packing
e. Whether first article inspection is required

6.3 Subject term (key word) listing.

Attachment
Tie-down

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – AS
Air Force – 11
DLA – GS

Preparing activity:
Navy – AS

(Project 1680-2022-001)

Review activity:
Navy – CG

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